

19 FOUNDATIONS IN COLD REGIONS

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19.1 INTRODUCTION

The design of foundations in cold regions differs significantly from that in temperate regions. Cold regions include both those areas with seasonal frost and perennially frozen ground (permafrost). On the basis of air temperature, snow depth, ice covers and permafrost, Bates and Bilello (1966) reported that about 48 percent of the northern hemisphere's land mass is categorized as cold regions and the southern most reaches of discontinuous permafrost over land masses approximately follow the 40°N latitude line, as illustrated in Figure 19.1. In cold regions, the upper soil layer, or active layer, experiences cycles of winter freezing and summer thawing.

In this region, the depth of the frost and thaw penetration varies widely depending on the air temperature, surface characteristics, and soil type and composition, including water content and thermal properties. These cyclical phase changes

a8 Pa

aw?" Limit of Cold Regions ~ FEE Continuous Permafrost (110) Discontinuous Permafrost

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Fig. 19.1 Boundaries of cold regions.

cause phenomena such as frost heave, downdrag force, and thaw settlement, which may cause foundation movement and seriously damage superstructures. Thus, the designer of foundations in cold regions must understand the physical, mechanical, and thermal properties of frozen soils. In addition, an evaluation of ambient temperature changes, soil deformation under freezing and thawing conditions, and long-term strength of foundation soils under different ground temperatures must be considered for the design life of a structure. The designer must ensure that a foundation meets the basic requirements for adequate depth, acceptable settlements and stability against failure. The basic foundation design concepts, such as allowable soil pressure and limiting short- and long-term settlement, that are applied to structure stability on unfrozen soils are also applicable to frozen soils. However, the design approach to foundations in frozen soils is more complex. Factors such as the ground's thermal regime, ice composition in frozen soils, load-deformation characteristics and creep behavior of foundation frozen soils, whether the structure is heated or unheated, construction schedule, and environmental constraints will govern the design and construction of foundations in frozen soils. The fundamentals of frozen-ground engineering are found in Phukan (1985).

The main purpose of this chapter is to present the fundamental aspects of foundation design in cold regions and the chapter is devoted to a discussion of frozen soils and temperature profile, design approach and considerations, thermal analysis, bearing capacity, and long-term settlement of shallow foundations and pile foundations.

19.2 FROZEN SOILS: PHASES AND TEMPERATURE PROFILE

Frozen soils consist of four phases—solid particles, unfrozen water, ice, and gas. As illustrated in Figure 19.2, the interphase relationships between soil particles and ice, ice and unfrozen water, unfrozen water and gas, unfrozen water and solid particles, and ice and gas are of prime importance and each constituent depends on the properties of each phase as well as external influences such as stress and temperature. Many soil mechanics texts (Lambe and Whitman, 1979; Holtz and Kovacs, 1981; and Chapter 3) have presented the physical relationships among the mineral, unfrozen water, and air components. The unfrozen water and ice phase composition, interface characteristics, and applicable thermodynamic relationship are more complex and are discussed by Anderson and Morgenstern (1973). The physical and mechanical properties of frozen soils are highly dependent on temperature and typical values of these properties are presented in Table 19.1. As the temperature is

735

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GAS

UNFROZEN WATER

ICE

SOLID PARTICLES

Fig. 19.2 Four-phase frozen soils.

lowered, the frozen soils become stronger and they become a three-phase material, of solid particles, ice, and gas. The relationship between the amount of unfrozen water and temperature is illustrated in Figure 19.3. Anderson and Tice (1973) have found that

$$W_{u,0} = 20^\circ \quad (19.1)$$

where

$W_{u,0}$ = unfrozen water content at 0°C , α = characteristics of soil parameter (Table 19.2) 0° = degree below freezing, $^\circ\text{C}$

The presence of salt solutions will change the general trend presented in Figure 19.3 and this shift may be compared with the freezing point depression of the soil solution.

The behavior of frozen soils under stress is highly dependent on the composition of ice and its temperature. Ice is a highly plastic material and its properties depend strongly on the internal below-freezing temperature, and the duration and magnitude of load applied. As such, the ice composition in frozen soils, which is described in terms of nonvisible, visible,

140 r —r 7 - r : r A. WYOMING BENTONITE, NATURAL r B. KAOLINITE
4 C, RUST (Fe,Os) ero D. LIMONITE 4 E,W. LEBANON GRAVEL <100u F.
FAIRBANKS SILT 1.00 + 4 ° TM“ o N = 4 o & 0.80 F 4 hs 9 = L 4 z= we S
o6of 4 c bs z > war 4 0.40 - ”4

to} -2 -4 -6 -8 -10 TEMPERATURE, °C

Fig. 19.3 Phase composition curves for six representative soils

and soil constituents. (After Anderson and Morgenstern, 1973.)

and silt ice, must be known for effective foundation design (Phukan, 1985). Illustrations of ice descriptions are presented in Figure 19.4.

The thickness of frozen soils is generally obtained by plotting the annual temperature variation in the ground. Ground temperature fluctuates with air temperature as shown in Figure 19.5. The temperature tends to vary seasonally with depth to 30 to 50 ft (10 to 15m). Below this depth of zero amplitude, the temperature generally increases under the geothermal gradient (average value of approximately 1.0°F per 100 ft (1°C in 30 m)) and depends on climate, geology, and geographic factors.

TABLE 19.1 MECHANICAL PROPERTIES OF VARIOUS FROZEN SOILS (Short-Term Condition).

STF 30°F 28°F 25°F 20°F Temperature (55°C) (—1.19°C) (—222°C) (—3.88°C)
(—666°C) Fairbanks Silt Density 1.5g/cm? Compressive strength, MN/m? 09
12 1.7 25 3.2 Tensile strength, MN/m? 0.2 0.8 1.2 1.8 2.3 Clay Density 1.70.
g/cm? Compressive strength, MN/m? 1.4 2.0 2.8 3.5 Fine Sand Density 1.90
g/cm? Compressive strength, MN/m? 18 3.5 6.0 8.0 Silty Sand Density 1.80
g/m? Compressive strength, MN/m?* 0.9 18 3.0 5.0

TABLE 19.2 SOIL PARAMETERS a AND f.

Soil Type ax 107? B

Limonite 8.81 — 0.33 Kaolinite 23.80 — 0.36 Wyomina bentonite 55.90 — 0.29
Fairbanks silt 48 — 0.33

a. NON-VISIBLE ICE

INCHES

Nbn Nbe WELL BONDED WELL BONDED NO EXCESS ICE EXCESS ICE
POORLY BONDED

b. VISIBLE ICE

° 7 — Q > f |S p RG ye ri ED \$ 3 +z 4 a 5 a ‘) Vx Ve Vr Vs INDIVIDUAL
ICE ICE COATINGS RANOOM OR STRATIFIED OR INCLUSIONS ON
PARTICLES IRREGULARLY DISTINCTLY ORIENTED ICE ORIENTED
ICE FORMATIONS FORMATIONS

c. VISIBLE ICE- GREATER THAN | INCH THICK

° | ie 33 = 4 5 6 ICE & SOIL ICE ICE WITH SOIL ICE WITHOUT SOIL
INCLUSIONS: INCLUSIONS LEGEND: soiL-[_] ice-EEg or G5) ee

Fig. 19.4 Ice description.

As illustrated in Figure 19.5 (which is generally described as the “whiplash” curve), the thickness of frozen soils, whether seasonal or perennial, is determined from the temperature distribution. Seasonally frozen soils undergo cycles of winter freezing and summer thawing.

The terms perennially frozen soils or permafrost both describe ground thermal conditions that remain below 32°F (0°C) continuously for more than 2 years. Permafrost distribu- tion in the world’s cold regions is well known (Pewe, 1966; Brown, 1967; Baranov and Kudryavtsev, 1966). Factors such as climate, geol- ogy, hydrology, topography, and biology determine the permafrost distribution. The permafrost regions are further divided into two zones (Figure 19.1): a “con- tinuous

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zone” in the north, followed by a “discontinuous” zone further south.

In continuous permafrost zones, perennially frozen soils exist everywhere beneath the ground surface except in newly deposited, unconsolidated sediments where the climate has just begun to impose its influence on the ground surface. The thickness of continuous permafrost varies from about 200 to 300 ft (60 to 90 m) at the southern limit of the continuous zone, possibly increasing to 3000 ft (1000 m) in the northern part of the zone in Alaska and Canada. The temper- ature of the permafrost in the continuous zone at the depth of zero amplitude varies from 23°F (— 5°C) in the south to about 5°F (— 15°C) in the extreme north.

In discontinuous zones, some areas have perennially frozen soils beneath the surface and other areas are permafrost free. The temperature at the level of zero amplitude in the discontinuous zone varies from a few tenths of a degree below 32°F (0°C) at the southern limit to 23°F (—5°C) at the boundary with the continuous zone. Permafrost is also known to occur in many northern arctic coasts and the permafrost thickness under subsea conditions may vary widely. Offshore permafrost is beyond the scope of this chapter.

In continuous or discontinuous frozen soils, foundations will be within the depth of zero amplitude, and the ground temperature fluctuations must be determined. The temperature regime of frozen soils depends on present and past climate, terrain factors, and the complex energy exchange between the ground and the

atmosphere. All aspects of foundations in cold regions are sensitive to heat transfer effects. Either a theoretical approach or actual ground temperature monitoring devices may be used to determine the temperature profile needed for the foundation design.

The warmest frozen ground temperature is given by

$$T_y = 32 - A_g + A_p \exp(-d/\alpha P) \quad (19.2) \text{ where}$$

T_y = temperature ($^{\circ}\text{F}$) at depth d A_y = amplitude of temperature difference between top of permafrost (32°F) and depth at which there is no temperature fluctuation α = thermal diffusivity of frozen soils P = a period of 365 days

Because there is a time lag between the air temperature and the ground temperature, the temperature profile calculated by Equation 19.2 is conservative. A time lag of approximately 3 months generally exists between the air temperature and the ground temperatures.

19.3. DESIGN APPROACH

In areas where there is no potential for thermal degradation, frozen soils with temperature well below freezing provide an excellent foundation support. In extreme cases, ice-rich frozen soils with near below-freezing temperatures may undergo extreme deformation even at low design load. Therefore, the design approach in frozen soils depends on the nature and composition of the frozen ground, ground temperature, type of structure to be founded and whether the structure will be heated or unheated, the magnitude of loads, etc. The following are the most common design approaches applied to the design of foundations in cold regions:

- a. Maintaining frozen ground in its frozen state
- b. Allowing frozen soils to undergo thawing caused by construction :
- c. Improving the site condition before construction
- d, Using conventional designs appropriate to unfrozen soils

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————— PERMAFROST 32

-30 -10 20

MEAN ANNUAL SOIL SURFACE TEMPERATURE)

50 TEMPERATURE $^{\circ}\text{F}$ -20 ile}

NON-PERMAFROST ———>

ow id

40 70 100

ENVELOPE OF ACTIVE MINIMUM GROUND LAYER TEMPERATURES
 |
 GROUND TEMPERATURES WHICH WOULD EXIST IN EARLY SPRING
 | (NOT TO SCALE)
 PERMAFROST
 AVERAGE GROUND A TEMPERATURE INCREASE _ AT |.8°F PER 100
 FT.
 DEPTH (NOT TO SCALE)
 SEASONAL FROST

Fig. 19.5 Whiplash curve.

Method (a) is a passive method commonly applied in thaw- unstable materials. The method is particularly applicable in ice-rich frozen ground where significant potential for deformation as a result of thermal degradation exists. Self-refrigerated systems or two-phase heat tubes, mechanical refrigeration systems, and a combination of a gravel pad with insulation and a thermal heat removal device may be used to retain frozen soils in their frozen state.

Method (b) is suitable for thaw-stable material where the anticipated total settlement or differential settlement due to change in thermal regime will not exceed the acceptable limits of the structure. This active method is used mostly in competent material. Extensive site investigation, thaw-settlement tests, and predictions are needed for the application of this method.

Method (c) is applied primarily to seasonal frost or dis- continuous permafrost areas where the thickness of the frozen soil layer is shallow enough to allow excavation of unsuitable material and replacement with competent thaw-stable material. Other site soil improvement techniques may involve prethawing the existing frozen ground and never allowing it to refreeze. Prethawing techniques, such as injection of steam or cold water, stripping of insulating surficial soils, and use of ground cover to increase solar heat input into frozen soils, may be applied to preclude the anticipated thaw settlement. However, problems such as metastability of soils, unworkability of ground surface, excessive pore pressure, or poor drainage may arise during the prethawing process and the designer must evaluate such detrimental effects before using this method.

Method (d) is only used when the supporting material consists of either ice-free competent bedrock or dense, thaw- stable sands and gravels. Depending on the magnitude of

structural loads, either shallow or deep foundations are installed using conventional allowable soil pressure theory.

19.4 DESIGN CONSIDERATIONS

Two basic considerations for foundation design in cold regions are stress effects on the supporting material and thermal phase effects between the structure and the foundation material. Stress effects include deformation and flow of material under the applied stress. Thermal phase effects encompass the heat flow between and thermal analysis of the structure and the supporting soils.

The designer must understand the behavior of frozen soils under different temperatures, confining pressures, and strain rates. Many publications (Tsytoich, 1975; Andersland and Anderson, 1978; Phukan, 1985) have presented the details of physical and mechanical properties of frozen soils. The designer must take into consideration the time-dependent as well as temperature-dependent behavior of frozen soils.

19.4.1 Creep Parameter and Long-Term Strength

Generally, the design of foundations in frozen soils is governed by the long-term strength. Creep parameters are determined from laboratory tests to define the long-term strength. As shown in Figure 19.6, the creep or time-dependent deformation behavior of frozen soils may result from (a) pressure melting of ice at points of soil-grain contact, (b) readjustment of the particles, (c) plastic deformation of pore ice, or (d) breakdown

Basic creep curve

Tertiary creep

1 1 ' Secondary creep 9 1/ Ice-rich N71 frozen ! \ !

; soils ! 1 \

Primory creep

Axial strain, ϵ

Ice- poor frozen soils

| L ' '

Time, t

Fig. 19.6 Creep behavior.

of structural bonds between ice and soil grain. The deformation rate goes through three stages—primary, secondary, and tertiary—before the failure of frozen soils. Deformation during the primary stage is calculated from the following relationship (Vyalov, 1962; Assure, 1963):

$$\epsilon = \epsilon_0 + \frac{\sigma}{m} t^m \quad (19.3)$$

ϵ = primary strain

σ = applied stress

t = elapsed time of application of load @, K, A, m = creep parameters

θ = degree below freezing, °C

where

Typical values of creep parameters are given in Table 19.3. The creep equation in the secondary state (Hult, 1966) may

be written as $\epsilon = E + a(\theta) + ea(2) \quad (19.4) \quad E \text{ ra } G,$

E = the Young's modulus

ϵ_0 = arbitrary small strain for normalization

ϵ_0 = arbitrary small strain for normalization θ_0 , θ_0 = temperature-dependent creep parameters

where

Equations 19.3 and 19.4 are used to determine the long-term deformation of frozen soils. Generally, ice-poor frozen soils show distinct primary creep behavior, whereas ice-rich frozen soil behavior is dominated by secondary creep. The long-term strength (σ_{∞}) of frozen soils is given by

B

$\sim \log.(1/B) \quad \text{“} _$

Fun

TABLE 19.3 CREEP PARAMETERS.

Lf)

Soil Type	m	A	K	psi(hr)/°C*	psi(hr)/°F*	Ottawasand	0.78	0.35	0.97	44.72
5500 Manchester										

fine sand	0.38	0.24	0.97	2.20	285	Suffield
-----------	------	------	------	------	-----	----------

clay	0.42	0.14	1.00	0.73	93	Sandy silt	0.27	0.10	0.89	0.88	90
------	------	------	------	------	----	------------	------	------	------	------	----

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where θ and B are parameters depending on soil type, its properties, and temperature and are determined from creep tests.

19.4.2. Thermal Analysis

The designer must perform an analysis of thermal behavior of the structure, ground and their interface. The basic heat transfer due to conduction, convection, and radiation must be understood through heat-flow calculations (Carslaw and Jaeger, 1959; Lunardini, 1981). For most engineering problems, conduction is the dominant heat-transfer process and its relationship to dry unit weight (γ_d) and water content is presented in Figures 19.7 and 19.8. The governing equation for steady-state conditions with no phase change is given by

= s27° oT e°T = =a (19.6) ct oz 160 T T —T T T T T © S, 100 %,DE-
GREE OF SATURATION 130 NOTE: 4 THERMAL CONDUCTIVITY K IS
EXPRESSED IN Btu PER HOUR PER 140 FT2 PER UNIT, THERMAL 4 x
GRADIENT IN OEG, F PER FT is (B/FT. HR. F) a ~ a cs 4 __ = 9 Pe wa z
3c = = a 2 > 4 ao 4 a » , 1 ‘ 4 1 1 1 1 IS 20 25 30 35 40 45

“W, MOISTURE CONTENT, (%)

(A) COARSE GRAINED SOILS— FROZEN esr 100%, DEGREE OF SATU-
RATION

130

T T T T T

UNIT WEIGHT, (LB/FT>)

Ya, DRY

i 10 i 20 25 30 35 40 45 “W, MOISTURE CONTENT, (%)

(B) FINE GRAINED SOILS— FROZEN

Fig.19.7 Thermal conductivity of frozen soils. (A) Coarse-grained soils. (B)
Fine-grained soils. (After Kersten, 1949.)

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160 T

T T T T T T ks Sp 100% DEGREE OF SATURATION

ORY UNIT WEIGHT (L8S/FT>)

Vas

° 5 10 15 20 «625 30 35 40 (45 “W, MOISTURE CONTENT, (%)o)

(A) COARSE GRAINED SOILS- UNFROZEN eos 100% DEGREE OF SATU-
RATION

130 Ca tf !20r ~ ao o =) — Or ise = 2 we = wor E 2 > 390 = > = r= 5 80 7Or
60

o 5 © i§ 2 2 30 35 40 45 “W, MOISTURE CONTENT, (%)

(B) FINE GRAINED SOILS-UNFROZEN

Fig. 19.8 Thermal conductivity of unfrozen soils, (A) Coarse- grained soils. (B)
Fine-grained soils. (After Kersten, 1949.)

where

T = temperature

t = time

z= depth

α = thermal diffusivity = k/c (k and c are thermal conductivity and volumetric heat capacity, respectively)

Equation 19.6 is rewritten for two dimensions as

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \quad (19.7)$$

Equation 19.7 is the Laplace equation and is used to solve various engineering problems using boundary conditions.

The steady temperature beneath heated or cooled areas on the ground surface is given by

$$T_z - T_\infty = \frac{T_s - T_\infty}{2} \left[1 - \frac{z}{\sqrt{z^2 + x^2}} \right] \quad (19.8)$$

where

T_z = desired temperature
 T_∞ = mean ground surface temperature outside heated or cooled area

z = depth

x = horizontal direction
 Equation 19.8 may be rewritten to determine the temperature regime under a long strip (such as a road of width $2a$) on the ground surface as

$$T_z - T_\infty = \frac{T_s - T_\infty}{2} \left[1 - \frac{z}{\sqrt{z^2 + a^2}} \right]$$

$$(19.9)$$

19.4.3 Frost Penetration and Depth of Thaw

Determination of frost depth or thaw penetration is one of the most important thermal estimations in cold regions. Such calculations are required to estimate change and depth of the active layer, frost or uplift force, and downdrag force. The placement of shallow foundations or determination of effective pile depth is governed by such estimations. The frost or thaw depth (x) is given by

$$x = \frac{A}{K} \left(\frac{I_a}{I_s} \right)^{0.5}$$

$$(19.10)$$

for U.S. customary units, and

$$x = \frac{A}{K} \left(\frac{I_a}{I_s} \right)^{0.5}$$

A = thermal constant, a function of thermal ratio (α) and fusion parameter (λ) (Fig. 19.9)
 K = average thermal conductivity for unfrozen and frozen soils in BTU /ft-hr-°F
 K_s = average thermal conductivity for unfrozen and frozen soils in J/S:m-°C
 n = the ratio between the air freezing or thawing index and surface freezing or thawing index (see Table 19.4 for typical values)
 I_a = air freezing

index, degree-days. I_a = air thawing index, degree-days. L = latent heat of fusion of soil, BTU/ft³ or J/m³

(19.11)

for S.L, where

EXAMPLE 19.1

Determine the depth of thaw penetration into a homogeneous frozen silt for the following conditions:

« Mean annual air temperature (MAT) = 36°F

e Air thawing index, J_a = 2500 degree-days

« Surface covered with snow

e Length of thawing season, t = 150 days

® Soil properties: dry unit weight density, γ_d = 85 pcf water content (w) = 30 percent

Solution

Volumetric latent heat and fusion: $L = 144$,

$L = 144 (85) (0.3) = 3672 \text{ BTU/ft}^3$

Average volumetric head, C , = $7(0.17 + 0.75)$

C , = $100[0.17 + (0.75)(0.30)] = 39.5 \text{ BTU/ft}^2\text{-}^\circ\text{F}$ Average thermal conductivity, K , = $4(K_u + k_f)$, where K_u = unfrozen thermal conductivity = $0.66 \text{ BTU/ft-hr-}^\circ\text{F}$ (Fig. 19.8)

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h COEFFICIENT

$T_1 T_2 = T_1 T_2 T_1 T_2$

03 oe' eT oer 1 0.01 0.02 0.03 0.05 0.07 0.10 0.12 0.15 0.20 0.30 0.50 0.70 1.00
FUSION PARAMETER, A & Fig. 19.9 A-values.

TABLE 19.4 n FACTORS.

Surface Freezing Condition Thawing Condition Snow 1.0 = Turf 0.5 1.0 Sand and gravel 0.9 1.5 Spruce 0.73 0.44 Willows = 0.82 Weeds — 0.86 Concrete road 0.8 2.03 Asphalt road 0.8 1.61 White painted surface — 1.12 K_f = frozen thermal conductivity = $0.94 \text{ BTU/ft-hr-}^\circ\text{F}$ K , = $4(0.66 + 1.05) = 0.85 \text{ BTU/ft-hr-}^\circ\text{F}$

Assuming $n = 1$:

I_1 (25 Average surface temperature, V , = — = (2) (2500)

$150 = 16.66^\circ\text{F}$ Mean annual soil temperature, $\bar{T}_s$ = MAT — $32^\circ\text{F} = 36 - 32 = 4^\circ\text{F}$ s 0 4 Thermal ratio, a = — = — = 0.24 V 16.66 Fusion parameter,

$pp = V_c A / L = (16.66)(39.5) / 3672 = 0.18$ From Figure 19.9, $A = 0.92$. 4 Estimated depth of thaw, $x = A ma$

$7 (48)(85)(1)(2500) = (0.92) 3672$
 $= 48 \text{ ft}$

EXAMPLE 19.2

Determine the depth of frost penetration into a homogeneous deposit for the following conditions:

* Mean annual air temperature (MAT) = 2°C * Surface freezing index, nl , = 1300 degree-days
 * Length of freezing season, t = 140 days * Soil conditions: silty gravelly soils unit weight, $\gamma = 2000 \text{ kg/m}^3$ * water content = 8 percent 2000

* $l = 0.14 + 0.08$

Dry unit weight, $\gamma_d = 1851.85 \text{ kg/m}^3$

Solution

$L = (334)(1851.85)(0.08) = 49.5 \text{ MJ/m}^2$

$1851.85 = \frac{2000(1 - w)}{1 + w} = \frac{2000(1 - 0.08)}{1 + 0.08} = 1742.5$
 $MI/m^2 \cdot ^\circ\text{C} = 2.04 + 2.16 k = COT \text{ ENO} = 2.10 \text{ w/m} \cdot ^\circ\text{C}$ ni 1300 VW, $= -f = -928^\circ = Fag = 928^\circ\text{C}$ Vo = $2 - 0 = 2^\circ\text{C}$ a | $9.28 \text{ m} \cdot \text{yr} = (9.28)(1.78) = 16.5 \text{ m}$ as 7033

From Figure 19.9, $A = 0.88$. Estimated depth of frost penetration:

ga nf

$L (7200)(24)(2.10)(1300) \cdot \text{yr} = 2.7 \text{ m}$

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Air thawing index, degree-days

0) 500 1000 1500 ($^\circ\text{C}$) a a m_fto 1000 2000 3000 ($^\circ\text{F}$) ie) o ——— TrrtTtyii |

r as sof Ree

\$f 4 Se Ft dl ae = 20 er Sa 3 r 8 NS & 6 | 3.0F 10-- Fa 4ol

Fig. 19.10 Frost penetration under different conditions. ©, Gravel; O, sand; A, 24-inch (61-cm) gravel, sand; x, 24-inch (61-cm) gravel, 1-inch (2.54-cm) insulation, sand. @, White-painted mat, sand.

Phukan (1985) presented the determination of the depth of frost or thaw penetration for nonhomogeneous soil conditions, and some typical values of frost penetrations under different boundary conditions are illustrated in Figure 19.10.

19.5 SHALLOW FOUNDATIONS

Shallow foundations are usually chosen when competent subsoil conditions are encountered at shallow depths or when the superstructure's design load is not sufficient to require deep foundations. In cold regions, shallow foundations are

generally constructed so that frost or thaw penetration will not affect the supporting foundation material. In addition to basic requirements regarding bearing capacity and long-term settlement, shallow foundations must be designed to withstand the adverse effect of thermal balance between the structure and the ground. Depending on the nature of the structure and the type and composition of supporting soils, including their temperature, the design approaches presented in Section 19.3 are used to maintain the thermal stability of foundation soils. Typical shallow foundation designs are illustrated in Figures 19.11 to 19.16. These foundations may be grouped into the following six major categories:

- . Footings on gravel pad with or without insulation . Raft on self-refrigerated gravel pad
- . Footings or raft on ventilated gravel pad Footings or raft on mechanical refrigeration
- . Post and pad foundations
- . Ground sills on original ground or on gravel pad

AUuWkwre

19.5.1 Bearing Capacity of Frozen Ground

The classic theory of unfrozen soil-bearing capacity (presented in Chapter 4) may also be applied to frozen soils. The major difference is the use of long-term frozen soil strength based on the creep parameters presented in Equation 19.5, Also, the

aN

Floating Concrete — Floor

7 a SE Ee

a r “4 [- 7 5 — “ ° / : Speer tae Aaa or Gels ; Gravel Pad « LY Ca eae ee e 4
bts a > +! Permafrost

insulation Footings

Fig. 19.11 Footings on an insulated gravel pad (not to scale,)

aN

Floating Concrete Heal Tubes ec Floor

Insulation ~ | Footings

a eS TI .

7 7 Permafrost

Fig. 19.12 Footings on a self-refrigerated gravel pad with heat tubes (not to scale.)

ao

Forced Floating Concrete ; : Air Ducts ge Floor Insulation Sy Pili

= = a = =e e mee et = Pea e o e 5 q *: 1 lo*-s6 E -

WITTX 2 *.’s4 Gravel Pad 2-0-9. 2-165 1S @r ‘777

Permafrost

Fig. 19.13 Footings on a self-refrigerated gravel pad with forced air circulation (not to scale,)

Floating Concrete Floor Insulation

Permafrost

fo 7. 77, 7 Fig. 19.14 Footings on a reinforced gravel pad (not to scale.)

‘a Concrete Raft Insulation x

3 EES

“20.2 67> Gravel Pad!’ 2 : = pe an

7 Permafrost

Fig. 19.15 Raft on grade (not to scale.)

\$ Floor Girder 2 f t — 7

Im Air Gap

-- Wood, Reinforced Concrete, or Steel Post

a

Active Layer

Permatrost

Concrete or Wood Base

Fig. 19.16 Post and pad (not to scale.)

frozen soil shear strength is dominated by the cohesion (c) factor only. The bearing capacity of frozen soils is given by

$dap = C NE$, (19.12)

where

ec = cohesion = $a_{,,}/27$, = long-term strength of foundation frozen soil, given by Equation 19.5 N , = bearing capacity factor = 5.14 &. = geometry of foundation

Other investigators (Vyalov, 1959; Tsytovich, 1975) have published bearing-capacity values for typical soils and some of the recommended bearing capacity values are presented in Table 19.5, A few published data sources are available for

ice-rich frozen soils at 23°F to 32°F (— 5°C to 0°C) and an empirical relationship can be written as:

$$C \text{ (KPa)} = 35 + 28 T \text{ (19.13)}$$

where T is the absolute value below freezing in degrees Celsius.

EXAMPLE 19.3

A square footing 6 ft by 6 ft is placed at a depth of 5 ft (below the active layer) in a deep stratum of frozen silt. Assume that there is air space between the floor and the natural ground to maintain thermal equilibrium conditions. Determine the bearing capacity of the frozen soils.

Solution

Assume that the long-term strength of frozen silt is determined from creep tests and is given by

$$1816 \text{ Out in } t_y + 8.15$$

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Assume the design life of structure ϵ , is 30 years $= (30)(365)(24) = 2.628 \times 10^6$ hr
 $1816 \text{ Ou in}(t_y) + 8.15$ Bearing capacity $= C N. \epsilon$,:

$$N. 1 N, = 5.14; \quad \epsilon = 0 + \frac{4}{14} = 119: \quad \therefore \text{soe ey yg Bearing capacity} = (88.03/2)(5.14)(1.195) = 270.35 \text{ psi}$$

19.5.2 Long-Term Settlement

Based on the bearing capacity shallow foundations may be designed in competent frozen soils with colder temperatures. However, long-term settlement generally governs the design, as frozen soil behavior is time-dependent as well as temperature-dependent. Ice-rich frozen soils or frozen soils with temperature close to freezing are especially susceptible to long-term settlement even at low loads. The long-term settlement in frozen soils is calculated from Equations 19.3 and 19.4.

Long-term settlement may also occur if the foundation frozen soils are subjected to gradual thermal degradation. If, under the design conditions, frozen soils are allowed to thaw, the total settlement(s) must be calculated by

$$i = n$$

$$S = y_{eh} + S_a$$

$$i = 1$$

$$(19.14)$$

where

ϵ ; = thaw strain h , = thickness of frozen soil layer S , = consolidation settlement

The determination of consolidation settlement is complex, and one-dimensional linear theory (Morgenstern and Nixon, 1971) may be used for predicting settlement.

Typical thaw strain values for different soils are illustrated in Figure 19.17. EXAMPLE 19.4

Determine the long-term settlement for the square footing given in Example 19.3. The temperature at the depth of zero amplitude is 25°F. Assume the following frozen soil properties:

« Frozen dry unit of soils, γ_d , = 80 lb/ft³

e Thermal conductivity of frozen soils, k , = 1.0 BTU/ft-hr-°F e Volumetric heat capacity of frozen soils, C , = 28.0 BTU/

ft-hr-°F

TABLE 19.5 TYPICAL DESIGN BEARING CAPACITY VALUES (Ksf).

Temperature Soils	Pure Ice	Clayey	Silty	Sandy	Gravelly	Gravel & Organic
31 °F (— 55°C)	1.5	1.8	2.4	4.0	5.8	30°F (—1.11°C)
= 1.9 ke a 43 48 6.9 28°F (—2.22°C)	08	3.0	3.7	5.5	6.3	9.8 25°F (— 3.88°C)
4.2 5.0 6.8 8.0 12.0 20°F (— 6.66°C)	2.1	5.0	6.5	75	9.0	13.5

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60-

50;-

BS (2) T

Thaw strain, percent w to) T

Gravel-sand

0 | j i 1 nN J 800 4000 1200 1400 1600 1800 2000 Frozen dry density, kg/m³

Fig. 19.17 Thaw strain values.

Solution — k_f — (1.0)(24)

$q =$ —

oS TS = OST HA c; 28.0 say

The warmest temperature distribution in the frozen soils is obtained by Equation 19.2.

$T_y = 32 - A_g + A_p \exp(-d_z/aP) = 32 - 5 + S \exp(-d_z/(0.857)(365)) = 32 - 5 + S \exp(-0.1d)$

The temperature calculations at depth are given in Table 19.6.

The allowable soil pressure, $q_{ar} = q_{uy}/FS$

Load on 6 ft x 6 ft footing= (6 x 6) qu,

270.35 = (6)(6) — (144)

= 700 747 lb = 350 tons

Creep parameters assumed are $m = 0.27$, $A = 1.10$, $k = 0.89 @ = 90 \text{ psi} \cdot \text{hr}/^\circ\text{F}^*$
Design period, $t = 340 \text{ yr} = 2.628 \times 10^9 \text{ hr}$ for long-term settlement calculation
is presented in Table

The following example illustrates the calculation of thermal flux between the structure and the foundation frozen soils and the determination of gravel pad thicknesses commonly used for shallow foundations in frozen soils (Figures 19.11 to 19.16).

EXAMPLE 19.5

A building (20m x 100 m) is to be supported on a slab-on- grade with an average floor temperature of 21°C . Design a

TABLE 19.6 TEMPERATURE CALCULATIONS

AT DEPTH. Depth from

Depth from Base of To.

Ground surface, ft Footing, d, ft F 5 0 32.0 6 1 325 7 2 31.0 8 3 30.7 9 4 30.4 10
5 30.0 12 7 295 15 10 28.8

20 15 28.1 25 20 27.7

duct system beneath the floor to maintain the existing frozen ground condition at the site. Assume that the building site has the following soil properties and design data:

« Frozen silt, dry weight = 1450 kg/m^3 ; water content, $w = 25\%$

e Air freezing index at the site = 4000 degree-days

« Freezing season = 215 days

Solution

Step 1. Calculate resistance Resistance of concrete, $d = 0.15 \text{ m}$

a $2^\circ\text{C}/\text{W} = \text{LBW}/\text{me} \sim 208 \text{ m}^\circ\text{C}/\text{Resistance of insulation, } d = 0.15 \text{ p} = 2 ==$
 $= 2.63 \text{ m}^\circ\text{C}/\text{W} \sim 0.0057 \text{ s}$

Total resistance, $R = R_1 + R_2 = 2.72 \text{ m}^\circ\text{C}/\text{W}$ Step 2. Calculate depth of thaw

Thawing season = 365 — freezing season (days) = 365 — 215 = 150 days

Floor thawing index, $I_{ft} = (21 - 0) 150 = 3150 \text{ degree-days}$ Latent heat of gravel, $L = (333.7)(1920)(0.05) = 32.9 \text{ MJ/m}^3$ Depth of thaw,

T200A71,,\!!2 x=KR| (14 0”) -1] f

$$7200(24)(97)2(3150) \sqrt{2} = ? \text{ a} - (\text{ryt} \cdot (\text{as} \times | |$$

=2m Step 3. Determine heat to be removed from gravel pad Heat to be removed from gravel including sensible heat = (2)(32.87) + 1 percent = 70 MJ/m?

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TABLE 19.7 LONG-TERM SETTLEMENT CALCULATION.

Zone Temperature Vertical Strain Thickness Below Freezing, Stress at® Vm Deformation h, ft deg F psi =| A' = (ea) (h) (99.21)(2.628 x 408)010 1/0.27 2 0.5 99.21 4:7 [(90) (0.5 + 1)°8 8 = 0,89 2 0.8 60.00 = 0.76 0.15 3 1.25 36.75 = 0.006 0.018 3 1.60 23.12 = 0.0006 0.001 1.949 ft (Excessive) During freezing season, this heat must be removed over a where

period of 215 days: 70 MJ/m? (215)(24) hr

$$= 0.014 \text{ MJ/m}^* \cdot \text{hbr}$$

Average rate of flow =

24 W/m? Step 4. Average thawing index at the surface Lx? (32.87)(2)?

$$i = *7200 \text{ A}^2 \text{ K} - (7200)(24)(0.97)7(1) = 892 \text{ degree-days}$$

This thawing index must be compensated by an equal freezing index at the duct outlet to assure freeze-back.

Step 5. Average temperatures Average pad surface temperature at the outlet of duct

$$\text{freezing index} = 892 \text{ --}$$

$$= \text{--} = \text{--} 42^\circ \text{C length of freezing 215}$$

Inlet air duct temperature

$$\text{-- air freezing index}$$

~ length of freezing

$$= = \text{m} - 18.6^\circ \text{C 215}$$

$$\text{Average temperature rise in duct} = \text{--} 4.2 \text{ --} (\text{--} 18.6) =$$

14.4°C Step 6. Heat flow Average heat flow between floor and inlet duct air,

21 --(18.6) , == 14S Wm? 272 Las Wim Average heat flow between floor and outlet duct 21 -- (4.2) =" = 9.6 W/m? 272 ere Average heat flow between total floor and duct 4. x 185496 oa wm?

Total heat flow = 12 + 4= 16 W/m? Step 7. Calculation of the air velocity in the duct OLD = V AypC,Tr

$$Q = \text{total heat flow, W/m?}$$

$$L = \text{length of duct, m}$$

D = duct spacing, m

V = Velocity of duct air, m/s

A_c = cross-sectional area of duct, m²

ρ = air density (assume 1.33 kg/m³) C_p = specific heat of air at constant pressure (assume

1000 J/kg-°C)

ΔT = temperature rise in duct air, °C $LD = 2BD \frac{A_{p,CyTr}}{(16)(100)(1.5)}$

$\sim (0.073)(1.33)(1000)(14.4)$ (using duct dia = 1 ft = 305mm)

= 1.7 m/s

19.6 PILE FOUNDATIONS

Pile foundations are widely used in continuous and discontinuous permafrost regions to provide greater stability of the thermal region. In addition, they provide an effective support to transfer both heavy vertical and lateral loads to a depth where volume change, as well as reduction of shear strength due to temperature changes are minimal. In comparison to shallow foundations, pile foundations provide a more effective way to isolate structures from existing ground conditions and maintain thermal equilibrium. Seasonal frost heave and downdrag force due to thawing are eliminated by using an effective pile embedment length. The construction of foundations in poor foundation material, such as frozen, ice-rich, fine-grained soils, deep seasonal frost areas or frost-susceptible soils, is very difficult and pile foundations are the only alternative for such sensitive soil conditions. Frozen soil behavior depends on ice-matrix, temperature, rate of loading, etc. These factors must be considered in designing pile foundations in frozen soils (Phukan, 1977).

Some of the basic philosophy used in designing pile foundations in unfrozen soils (Chapter 13) may be applied in frozen ground. However, load-transfer mechanisms, installation techniques, and field pile loading tests are quite different. The load-carrying capacity of piles in cold regions is dependent on the adfreeze between the pile surface and the surrounding frozen natural soil or backfill material placed between the pile surface and the natural frozen ground. Figure 19.18 presents generalized load-transfer conditions at different times.

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Pile shaft stress, kPa 0 20 40 60 80 100 120 O T T T T

T T $t = 0.3 \text{ days}$ $t = 5 \text{ days}$ $t = 10 \text{ days}$ $t = 15 \text{ days}$ $t = 20 \text{ days}$

$t = 3 \text{ days}$ or $10 = 2 \text{ a}$

$t = 10 \text{ days}$ 15

1200 20

Fig. 19.18 Load transfer along pile shaft.

The state of the art of pile design has dramatically changed in the last decade owing to improvement in pile installation techniques and the use of a thermal device called the heat tube. Piles with thermal devices are called thermal piles and various configurations of heat tubes in thermal piles are illustrated in Figure 19.19. Depending on installation technique, piles are divided into three groups such as thermal piles, slurried piles and drilled-in-place piles. As explained earlier, the thermal piles may have different heat tube configurations as shown in Figure 19.19. The slurried piles do not have any heat tube and the annulus between the drilled hole and the pile surface is filled with slurry as illustrated in Figure 19.20. The drilled-in-place piles are generally driven into the soils and are most commonly installed in cases where a thawed soil profile is underlain by frozen soils.

In cold permafrost where frozen soils with colder temperature exist, the pile is placed in an oversized hole (generally by more than 0.5 ft or 0.152m) and the annulus between the pile and the hole is backfilled with a sand slurry. This pile is called a slurried pile (Fig. 19.20) and typical sand slurry gradation is given by

U.S. Standard Percent finer

sieve size by weight 9.52 mm 100 No. 4 93 to 100 No. 10 70 to 100 No. 40 15 to 57 No. 200 0 to 17

Thermal piles may be installed similarly to slurried piles, but heat tubes are added to offer greater long-term thermal balance. Thermal piles may also be placed by (a) direct driving into natural frozen ground, (b) driving into under-sized holes or steamed holes of diameter less than the pile diameter, (c) use of prethawed holes of diameter greater than the pile diameter. Thermal devices should be installed in piles placed in thaw- unstable frozen soils with temperatures close to freezing (Le., warmer frozen soils).

In North America, prefabricated timber or steel piles are most commonly used in cold regions, whereas in the U.S.S.R., precast, reinforced-concrete piles are installed. The major problem with concrete piles is their weakness against frost heave force, which produces tensile forces in the pile. Concrete has

fe

a

Ez Gs Thermal device Thermo pile inside pile ll Z A} = SS — = Thermal device Thermal device outside pile wrapped around pile

Fig. 19.19 Heat tube configuration.

Ground level

Acti ve layer

ee Pine pile

Slurry type Predrilled hole “eel

End plate Fig. 19.20 Slurried pile.

low tensile strength. Moreover, economic considerations dictate the use of either steel or timber piles.

19.6.1 Single Pile Load Capacity

Pile design for frozen soils must satisfy both thermal and stress—deformation (rheological) requirements as outlined in Section 19.4. In addition, the design must comply with the following.

- a. It should have an adequate factor of safety (at least 2) against gross failure.
- b. The settlement (both short-term and long-term) should be within the acceptable limits of the superstructure.
- c. There will be no unacceptable changes in the ground thermal regime due to the construction of the superstructure. Heat transmission by conduction down the pile must be prevented.
- d. Freeze-back of placed slurry must be achieved before loading of piles.

Pile design is generally based on the long-term adfreeze strength between the pile surface and the surrounding frozen material. Typical values of adfreeze strength at temperatures below freezing are presented in Table 19.8 and the vertical load capacity of typical circular pile is presented in Figure 19.21.

Generally, only end bearing is considered when piles are embedded in dense thaw-stable granular material or ice-free bedrock.

The designer must consider both summer and winter loading conditions to determine pile load capacity. The pile load capacity in summer conditions is given by

$$Q+W+P_s=P_s+P_b; \quad (19.15)$$

Assuming $P_s = 0$,

$$Q+W+2dL_a t_s = n_d L_t, \quad (19.16)$$

where

Q = structural load on pile W , = weight of pile

P_s = downdrag force

P_b = shaft resistance P_b = end bearing

d = diameter of pile

L_a = active layer thickness

L_t = effective embedment length of pile

t_s = shear strength of soil in the active layer (limited to 500 psf or 400 MPa) t_a = adfreeze strength of pile surface and soils
 Under winter conditions, the pile load capacity is given by $P_u = Q + W + \sum_{i=1}^n L_i t_{si}$
 or $P_u = Q + W + \sum_{i=1}^n L_i t_a$
 where P_u = uplift or frost heave force.
 (19.17)

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$D = 18$ in

Vertical load capacity, kips

$D = 6$ in

a. h. |

ie) 5 10 15 20 2 Embedment length below seasonal thaw depth (2), ft

Fig. 19.21 Pile load capacity.

If, during the winter, when active uplift force exists, the structure has not been built, Equation 19.17 becomes

$$P_u = W + \sum_{i=1}^n L_i t_a \quad (19.18)$$

The uplift force (P_u) is the most critical term in Equation 19.18 because of the absence of structural load on the pile. Some typical values of uplift force are:

Soil Type P_u ; Silty (most frost-susceptible) soils 40 psi (270 kPa) Organic soils 10 psi (70 kPa) Silty granular soils 20 psi (140 kPa)

Using Equations 19.16 and 19.18, the most critical effective embedment length of the pile against the structural load is determined. When the required effective embedment length is significant, because of uplift force, the techniques that should be used to mitigate frost heave force are:

- Isolation of the pile surface from the surrounding soils by wrapping with a suitable material such as visqueen
- The use of heat tubes in the active layer to reduce frost heave force (it is theorized that the radial heat flow causes ice lenses to grow parallel to the pile surface, resulting in no upward pressure on the pile surface)
- Replacement of frost-susceptible soils in the active layer with non-frost-susceptible soils

TABLE 19.8 TYPICAL VALUES OF ADFREEZE STRENGTH.

Temperature (°F)	Temperature (°C)	Adfreeze strength (psi)	Adfreeze strength (kPa)
39	5	5	34.5
30	-1.1	10	68.9
28	-2.2	15	103.4
25	-3.9	20	137.9
20	-6.7	25	172.4

psi (kPa)

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19.6.2 Single Pile Freeze-Back

The designer must calculate the freeze-back time required before allowing any load placement on pile installed in frozen ground. Either natural or artificial freeze-back may be used so that the installed pile regains adequate load capacity. The natural freeze-back method is used when sufficient cold reserves are available in the natural frozen ground. The best time for natural freeze-back is late spring when the ground temperature is lowest. In artificial freeze-back methods, the total heat that must be removed from the slurry is calculated and artificial refrigeration is used to remove the heat. Example 19.6 shows the step-by-step procedure for calculating refrigeration capacity requirements for artificial freeze-back.

EXAMPLE 19.6

A 12-inch diameter pile is placed in an 18-inch pre-augered, frozen, ice-rich silt deposit that is frozen well below the depth explored by drilling. A pile backfilled with slurry is installed to a depth of 35 ft. The following site conditions should be used to determine the pile load capacity and freeze-back time required.

PERMAFROST SILT:

Average warmest temperature = 30°F Dry unit weight = 85 pcf

Water content = 25 percent

Uplift force = 20 psi

e Active layer = 4 ft

SLURRY BACKFILL SAND-WATER

e Placement temperature = 40°F e Dry unit weight = 100 pcf

e Water content = 15 percent

e Slurry must be refrozen and cooled to 25°F

Solution

(a) SUMMER CONDITIONS

Assuming shear strength in active layer = 500 psf, Downdrag force, $P_d = 2(1)(4)(500) = 6280$ lb Shaft resistance, $P_s = 2(1)(35 - 4)(144)(5 \text{ psi})$

$= 70085$ lb (assumed t , at 30°F = 5 psi)

Neglecting the weight of the pile from Equation 19.16 $Q + P_d = P$, or $C = P_s - P_d$
 $= 70085 - 6280 = 63805$ lb Allowable pile load capacity in summer condition
 $= as = 31902$ lb

(b) WINTER CONDITIONS Uplift force, $P_u = (md L)(20 \text{ psi})$

$= m(1)(4)(144)(20) = 36173$ lb Neglecting the weight of the pile, from Equation 19.18, $P_y = 1d Lt$,

$n(1)(35 - 4)(144)(5) = 70085$ lb

Wil

The shaft resistance will resist the uplift force, therefore 70085

$2 = 35043$ lb Summer conditions govern the design.

allowable pile load =

(c) FREEZE-BACK CALCULATION Unfrozen volumetric heat capacity of the slurry is

$C_u = \gamma_a(0.17 + (1)@) = 100[0.17 + (1)(0.15)] = 32$ BTU/ft³-°F Frozen heat capacity of the slurry is $C_y = \gamma_u L[0.17 + (0.5)@] = 1000.17 + (0.5)(0.15) = 24.50$ BTU/ft³-°F Latent heat of slurry is $L = (144)(\gamma_4)@ = (144)(100)(0.15) = 2160$ BTU/ft³ Volume of slurry placed is $(0.75 - 0.5)(35) = 6.869$ ft³ Heat required to depress the slurry temperature to freezing point is $(C_u) \times (\text{volume of slurry}) \times (\text{temperature above freezing of slurry}) = (32)(6.869)(40 - 32) = 1758$ BTU Heat required to freeze the slurry is

$(L) \times (\text{volume of slurry}) = (2160)(6.869) = 14837$ BTU

Heat required to depress the slurry temperature from the freezing point to 25°F is

(e,) $\times (\text{volume of slurry}) \times (\text{required temperature below freezing})$

$= (24.50)(6.869)(32 - 25) = 1178$ BTU Hence, total heat to be abstracted from the slurry is $1758 + 14837 + 1178 = 17773$ BTU

Assuming the artificial freeze-back capacity to be 17 000 BTU /day, freeze-back time by artificial refrigeration is

$17773 = 1.045$ day or 25 hr

19.6.3 Long-Term Pile Displacement

In ice-poor frozen soils, the pile settlement may be limited by calculating the allowable adfreeze strength and then determining the depth of pile embedment needed to carry the designed load with a suitable factor of safety. These soils have grain-to-grain contact and the behavior is dominated by primary creep, as illustrated in Figure 19.6. If the design load is limited within the primary creep range, the long-term settlement may be limited. On the other hand, the load—deformation behavior of ice-rich soils is largely dominated by secondary creep (Fig. 19.6) and the properties of ice and the soil temperature are responsible for the long-term deformation. Accordingly, a creep flow load should be introduced to mathematically describe the pile behavior. The creep rate of ice-rich frozen soils may be written as

$\dot{\epsilon} = Br^n$ (19.19)

where

$\dot{\epsilon}$ = creep rate τ = level of shear stress B, n = creep parameters

Typical values of B and n for ice (Morgenstern et al., 1980) are:

$\dot{\epsilon}$, Temperature, °C kPa yr⁻¹ n — 1 45x 10⁻⁸ 3 — 2 2.0x 10⁻⁸ 3 — 5 1.0 x 10⁻⁸ 3 — 10 5.6 x 10⁻⁸ 3

If the constitutive behavior of material is characterized by Equation 19.19, the pile displacement rate ($\dot{u}_p$) under constant tangential shear stress (τ) and uniform ground temperature is given

by (Morgenstern et al., 1980) . $\dot{u}_p = \frac{2\tau a}{B} \left(\frac{\tau}{B} \right)^{1/n} (19.20)$ where a = pile radius. The long-term pile displacement is calculated by $u_p = \frac{2\tau a}{B} \left(\frac{\tau}{B} \right)^{1/n} \frac{t}{n+1}$

$n+1$

where t = design life of the structure. EXAMPLE 19.7

Determine the anticipated pile settlement in 30 years for a load of 17.5 tons in Example 19.6.

Solution

Tangential shear stress is

$\tau = \frac{17.5 \text{ tons}}{2 \times 10 \text{ ft}} = 0.875 \text{ tons/ft} = 0.875 \times 2000 \text{ lb/ft} = 1750 \text{ lb/ft} = 17.04 \text{ kPa}$ Assuming the creep parameters

$B = 17.04 \text{ kPa}$

$a = 3 \text{ ft} \quad B(25^\circ\text{F or } -3.88^\circ\text{C}) = 1.5 \times 10^{-8} \text{ kPa} \cdot \text{yr}^{-1}$

then from Equation 19.6, $\dot{u}_p = \frac{2 \times 3 \text{ ft} \times 17.04 \text{ kPa}}{1.5 \times 10^{-8} \text{ kPa} \cdot \text{yr}^{-1}} \left(\frac{17.04 \text{ kPa}}{1.5 \times 10^{-8} \text{ kPa} \cdot \text{yr}^{-1}} \right)^{1/4} (30 \text{ yr}) = 0.0015 \text{ m}$ (negligible in this case)

REFERENCES

Andersland, O. B. and Anderson, D. M. (eds.) (1978), Geotechnical Engineering for Cold Regions, McGraw-Hill Book Co., Inc., New York, N.Y.

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Anderson, D. M. and Morgenstern, N. R. (1973), Physics, chemistry, and mechanics of frozen ground, North American Contribution, Proceedings of the Second International Conference on Permafrost, Yakutsk, USSR, National Academy of Sciences, Washington, D.C., pp. 257-288.

Anderson, D. M. and Tice, A. R. (1973), The unfrozen interfacial phase in frozen soil systems, Ecological Studies, 4, pp. 197-125.

Assure, A. (1963), Discussion on creep of frozen soils, North American Contribution, Proceedings of the First International Conference on Permafrost, Lafayette, Indiana, National Academy of Sciences Publication 1287, pp. 339-340.

- Baranov, I. J. and Kudryavtsev, V. A. (1966), Permafrost in eurasia, Proceedings of the First International Conference on Permafrost, Lafayette, Indiana, National Academy of Sciences Publication 1287, pp. 98-102.
- Bates, R. E. and Billello, M. A. (1966), Defining the Cold Regions of the Northern Hemisphere, U.S. Army CRREL Technical Report 178.
- Brown, R. J. E. (1967), Permafrost in Canada, National Research Council, Canada, Map 1246A.
- Carlsaw, H. S. and Jaeger, J. C. (1959), Conduction of Heat in Solids, 2d ed., Oxford University Press, London.
- Di Pasquale, L., Gerlek, S. and Phukan, A. (1983), Design and construction of pile foundations in Yukon-Kuskokwim delta, Alaska, Fourth International Conference on Permafrost, Fairbanks.
- Holtz, R. D. and Kovacs, W. D. (1981), An Introduction to Geotechnical Engineering, Prentice-Hall, Inc., Englewood Cliffs, N.J., p. 733.
- Hult, J. A. H. (1966), Creep in Engineering Structures, Blaisdell, Waltham, Mass.
- Johnston, G. H. (1981), Permafrost: Engineering Design and Construction, John Wiley and Sons, Inc., New York, N.Y.
- Kersten, M. S. (1949), Thermal Properties of Soils, University of Minnesota, Engineering Experiment Station, Bulletin 28.
- Lambe, T. W. and Whitman, R. V. (1979), Soil Mechanics, John Wiley and Sons, Inc., New York, N.Y.
- Lunardini, V. J. (1981), Heat Transfer in Cold Climate, Van Nostrand Reinhold Co., New York, N.Y.
- Morgenstern, N. R. and Nixon, J. F. (1971), One-dimensional consolidation of thawing soils, Canadian Geotechnical Journal, 8, No. 4, pp. 448-565.
- Morgenstern, N. R., Roggensack, W. D., and Weaver, J. S. (1980), The behavior of friction piles in ice and ice-rich soils, Canadian Geotechnical Journal, 17, pp. 405-415.
- Pewe, T. L. (1966), Ice-wedges in Alaska: Classification, distribution, and climatic significance, Proceedings of the First International Conference on Permafrost, Lafayette, Indiana, National Academy of Sciences Publication 1287, pp. 76-81.
- Phukan, A. (1977), Pile foundation in frozen soils, ASME Energy Technology Conference, Houston, Texas.
- Phukan, A. (1980), Design of deep foundations in discontinuous permafrost, ASCE Spring Convention, Technical Session on Deep Foundations, Portland, Oregon.

Phukan, A. (1981), Design Guide for Roadways on Permafrost, State of Alaska, Department of Transportation and Public Facilities, Division of Planning and Program Research Section, Fairbanks.

Phukan, A. (1985), Frozen Ground Engineering, Prentice-Hall, Inc., Englewood Cliffs, NJ.

Tsyтович, N. A.(1975), The Mechanics of Frozen Ground, McGraw-Hill Book Co., Inc., New York, N.Y.

Vyalov, S. S. (1959), Rheological Properties and Bearings Capacity of Soils, U.S. Army CRREL, Translation 74 (1965).

Vyalov, S. S. (1962), The strength and creep of frozen soils and calculations for ice—soil retaining structures, U.S. Army CRREL, Translation 76 (1965).